

(Sheet metal work)

SD

When thickness of Engg material (metal) is 0.5^{mm} to 6 mm is considered as sheet.

↳ Thickness is less than 0.5 mm, it is called leaf or foil.

↳ " " more than 6 mm, it is called plate.

Sheet metal → thickness of material (metal) is less than 6 mm.

↳ Sheet metal are specified in term of gauge number, length and width.

↳ The thickness of sheet varies inversely proportional to the gauge number. i.e. higher the gauge number, the smaller is the thickness and vice-versa.

↳ Gauge Number can be known using standard wire gauge.

↳ Applications In automobile bodies, Almirah, heating ducts, domestic heating appliances, tubelight fixture, Container for ghee & oil, tanks, pans, trunks etc.

material used for sheet metal →

(1) Galvanised Iron → (G.I) →

- ↳ known as G.I sheet.
- ↳ These sheets are made using galvanisation process i.e. coating of zinc on the surface of soft iron. Zinc coating provides corrosion resistance surface and also improves the appearances.

(2) Copper → Copper is costlier than iron or steel but

It has certain specific advantages. Copper has

↳ High thermal & electrical conductivity.

↳ Useful strength with high ductility.

↳ Corrosion resistance.

→ Because of these properties, copper & its alloy are widely used in electrical application.

3) Stainless steel → It is alloy of Iron, Carbon, Chromium & Nickel. It is highly corrosion resistance and has bright surface without any coating.

4. Aluminium → Al has good workability, light weight, corrosion resistance and good electrical & thermal conductivity. The

③
specific weight of Al. is one third of the weight of steel for equivalent volume.

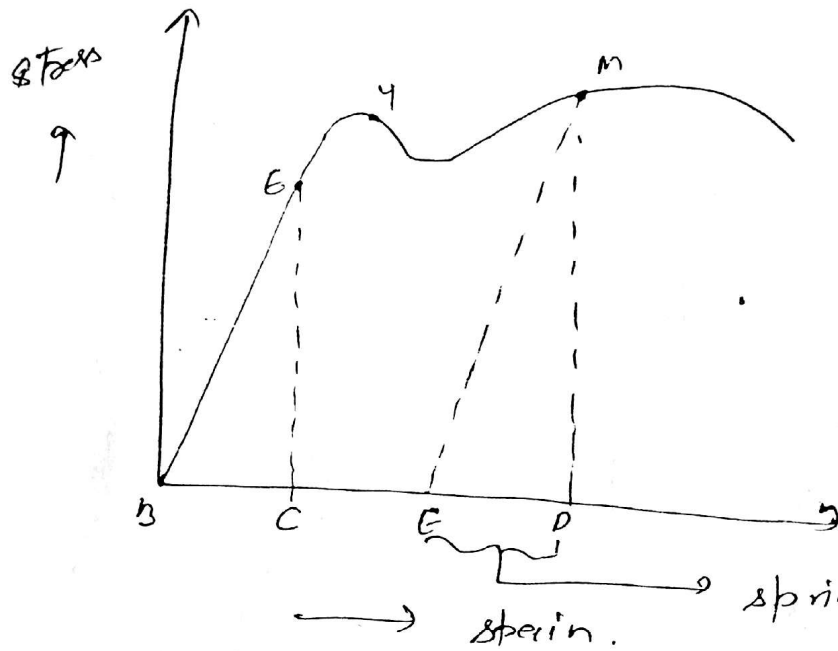
Tin → It is used as coating on Iron & Steel sheet to improve corrosion resistance. It provides resistance to corrosion and resists rusting. It is widely used in packet food containers, containers for edible oil & ghee, cans etc.

Lead → It is very weak, soft, malleable and low melting point metal. It possesses high resistance to corrosion. It finds application in radiation shielding sheet, inner lining for acid tanks etc.

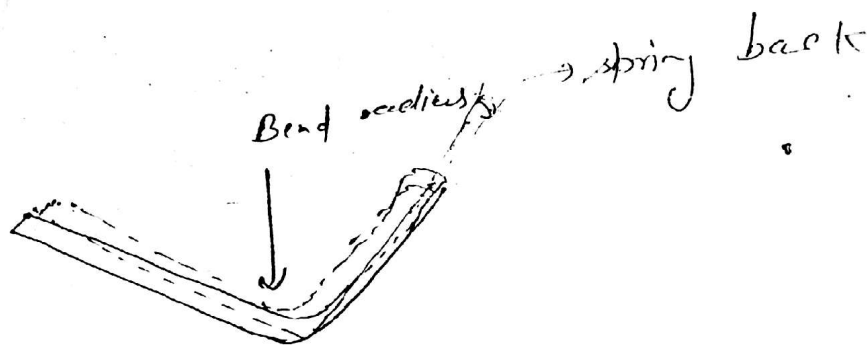
Spring back in sheet metal → In metal working

processes, when the metal is deformed, it is first elastically deformed & then plastically. When the load is removed, there is an elastic recovery by the metal and total deformation will get reduced a little. This phenomenon is called a spring back & is of more importance in operation such as bending.

↳ When the bending force is removed the workpiece moves back slightly towards its original shape as a result of elastic deformation.



BC - elastic deformation
CD - plastic strain
ED - spring back



sheet metal operations

① shearing process → cutting of sheet metal
in press working is a shear process.

↳ punch & die set are used for this.

↳ As the punch touches the metal & moves

downwards, it pushes the material into die opening. The material is subjected to both tensile

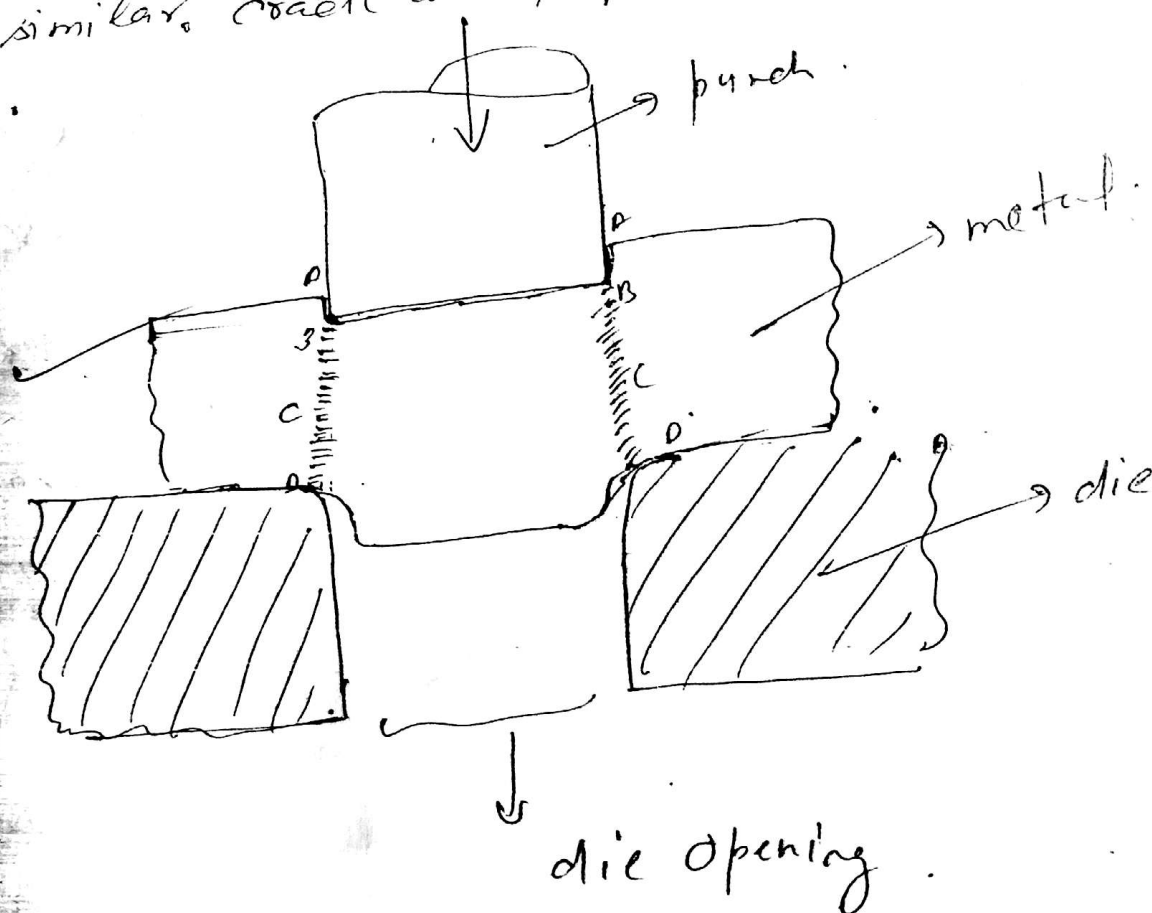
and compressive stresses.

When the local internal strain in the surface AB reaches its limiting value, the fibre ruptures.

Since the local strain is maximum at the corner, crack opens up just ahead of punch corner.

Thus fracture line BC propagates following a path along which successive inner fibre attains the fracture strain.

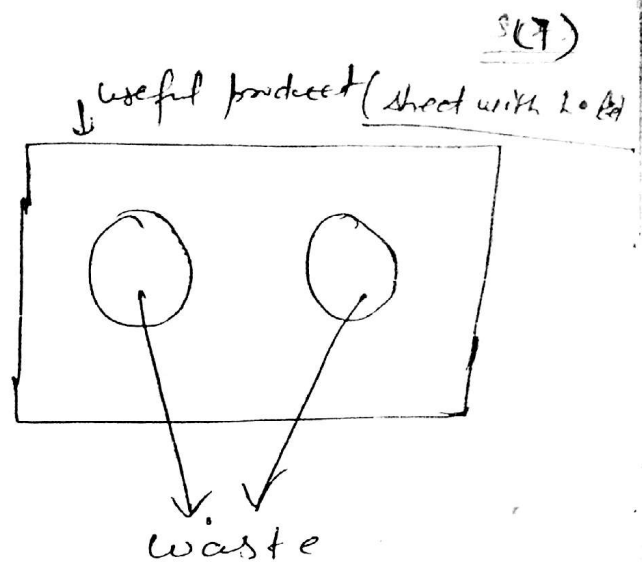
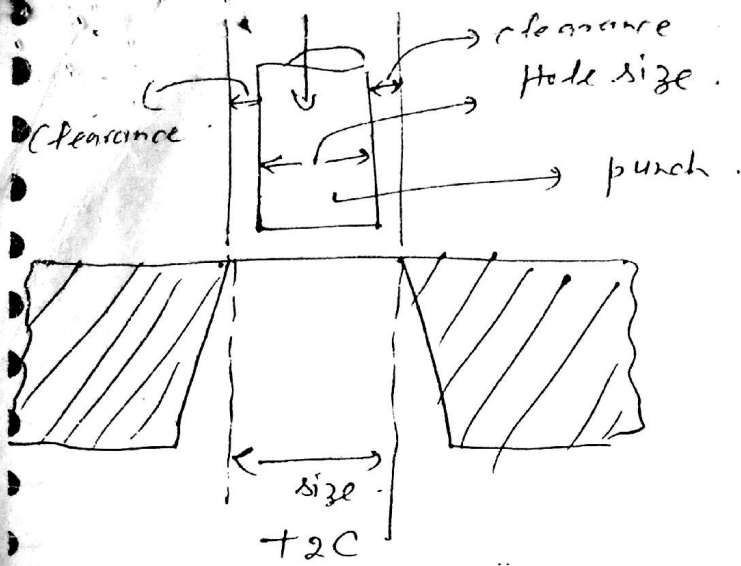
A similar crack also propagates from the die corner DE.



- (b) - Step 2
- ↳ Then crack starting from punch (BC) and die edge (DE) will meet and shearing complete, when the clearance b/w punch & die correct
 - ↳ If the clearance is too small, then the fracture lines do not meet & secondary deformation takes place
 - ↳ if the clearance is too large, fracture line do not meet and material is being dragged.
-

Punching → The punching is cutting operation by which various shaped holes are made in sheet metal strip.

- ↳ In punching operation, sheet with the hole is desired product and material punched out is ~~core~~ scrap.
- ↳ In this, punch size equals the size of hole & die opening size is obtained by adding the clearance to punch size.

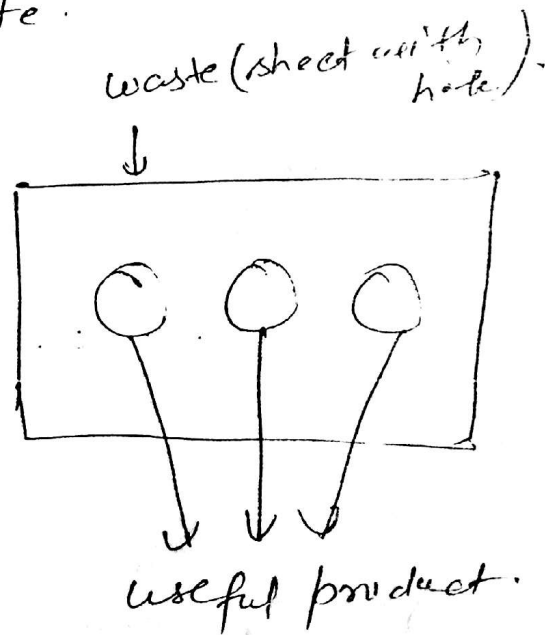
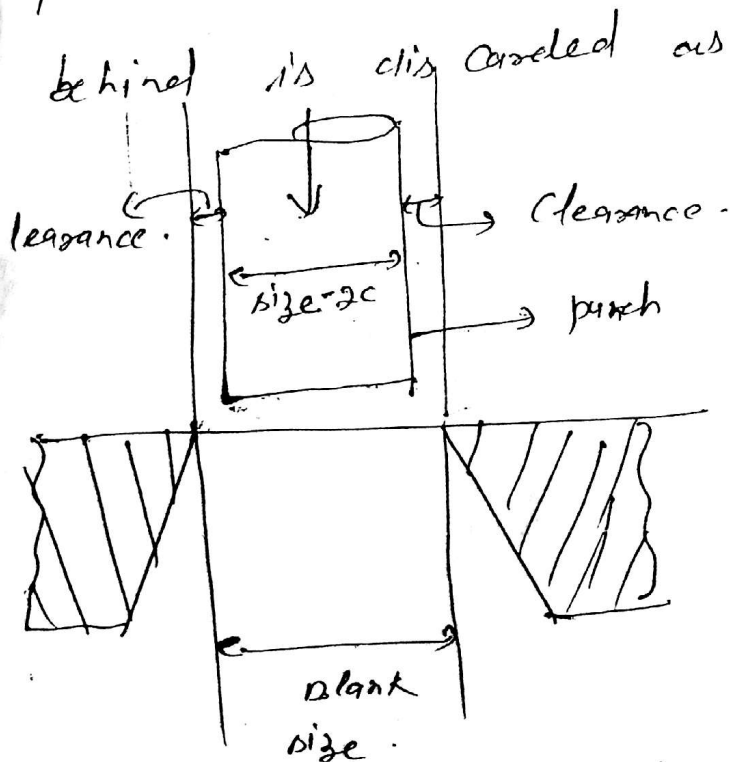


Hole size = punch size.

Hole size + 2x clearance = die opening size.

Blanking → is the operation of cutting out flat shapes from sheet metal strip.

The removed portion is called blank which is required product of the operation and metal with hole left behind is discarded as waste.



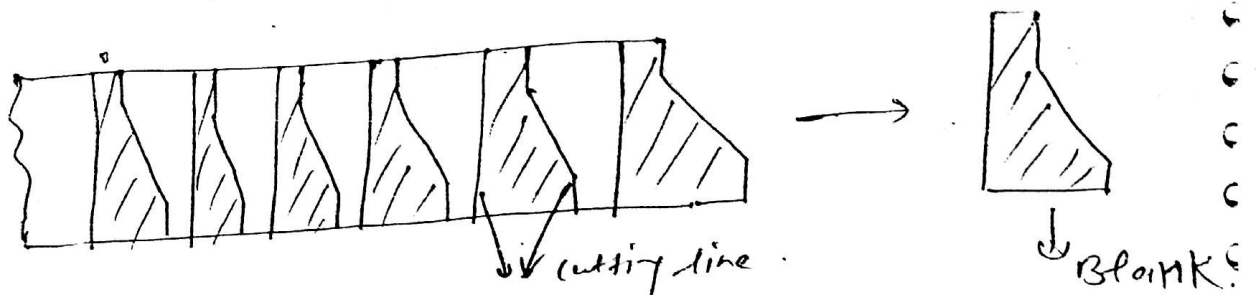
Blank size = die opening size.

Blank size + 2x clearance = punch size

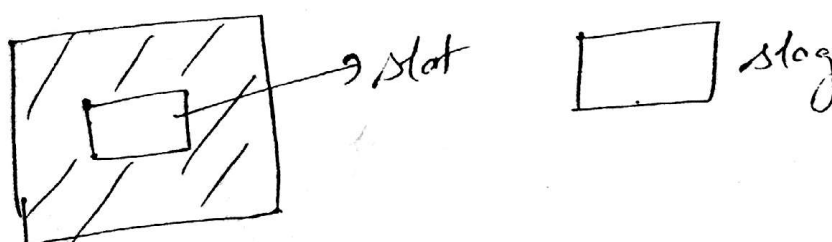
(8) ~~80~~ C
In this, die opening size equal to blank size and punch size is obtained by subtracting clearance from die opening size.

Cut-off & parting → To which blank are taken away from a sheet metal strip by cutting the opposite side of the part in a sequence.

↳ Typically, a new part is produced with each stroke of the punch and no-scraper is produced. (the blanks are nested with each other).



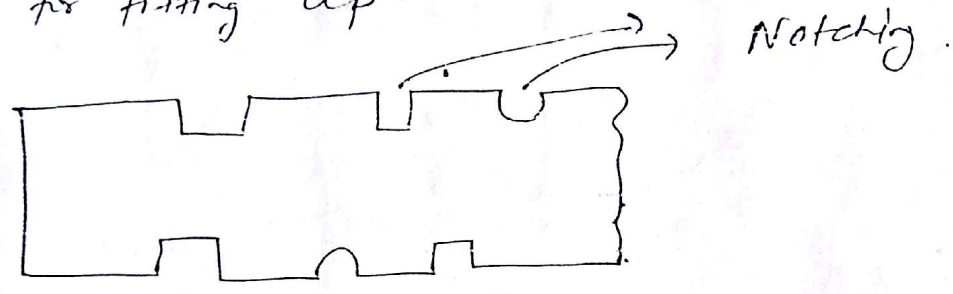
Slotting is slotting is a punching operation by which rectangular hole is cut out from the sheet metal.



Notching → It remove small portion of metal along of the edge of the stock.

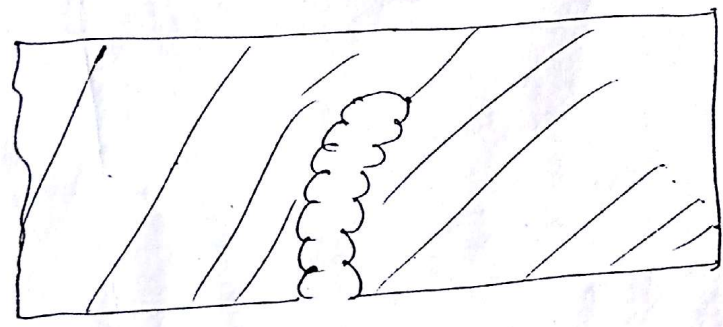
↳ A Notch can be made in any shape.

↳ The purpose of Notching is generally to release the metal for fitting up.

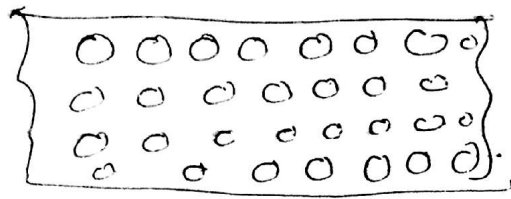


Rolling → In this, a specific contour is cut by producing a series of overlapping notches.

↳ The operation may be performed to produce any desired shape such as flange, collar etc.



Perforating → is an operation in which a number of uniformly spaced hole are punched in a sheet of metal. They cover the entire sheet of metal.



Bending → it is plastic deformation of metal about a

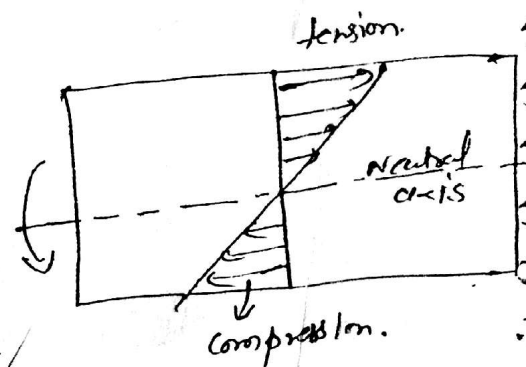
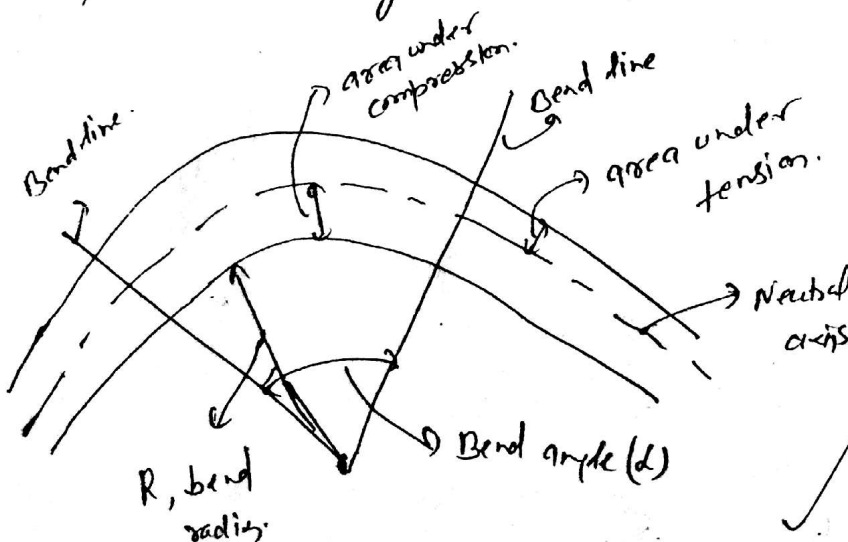
linear axis with little or no change in surface area.

↳ Due to applied forces, metal on the outside is stretched while on the inside is compressed.

↳ There is a plane in b/n which is neither stretched nor compressed. - This plane is known as neutral plane.

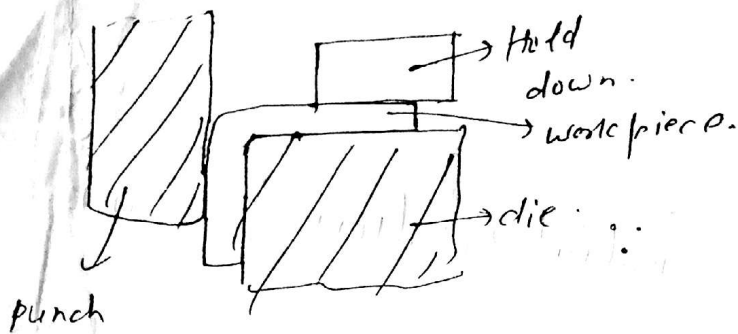
↳ For bending, there is some elastic recovery after the punch is removed. This is known as spring back.

↳ Thus, if metal is to be bent at specific angle the metal must be slightly over bent to compensate for unbending action of spring back.

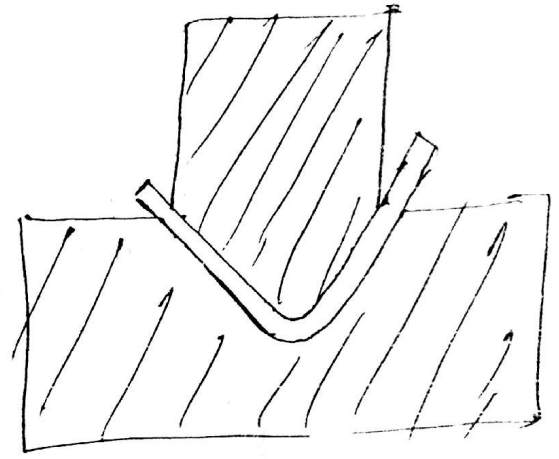


Different type of bending methods

(1b)



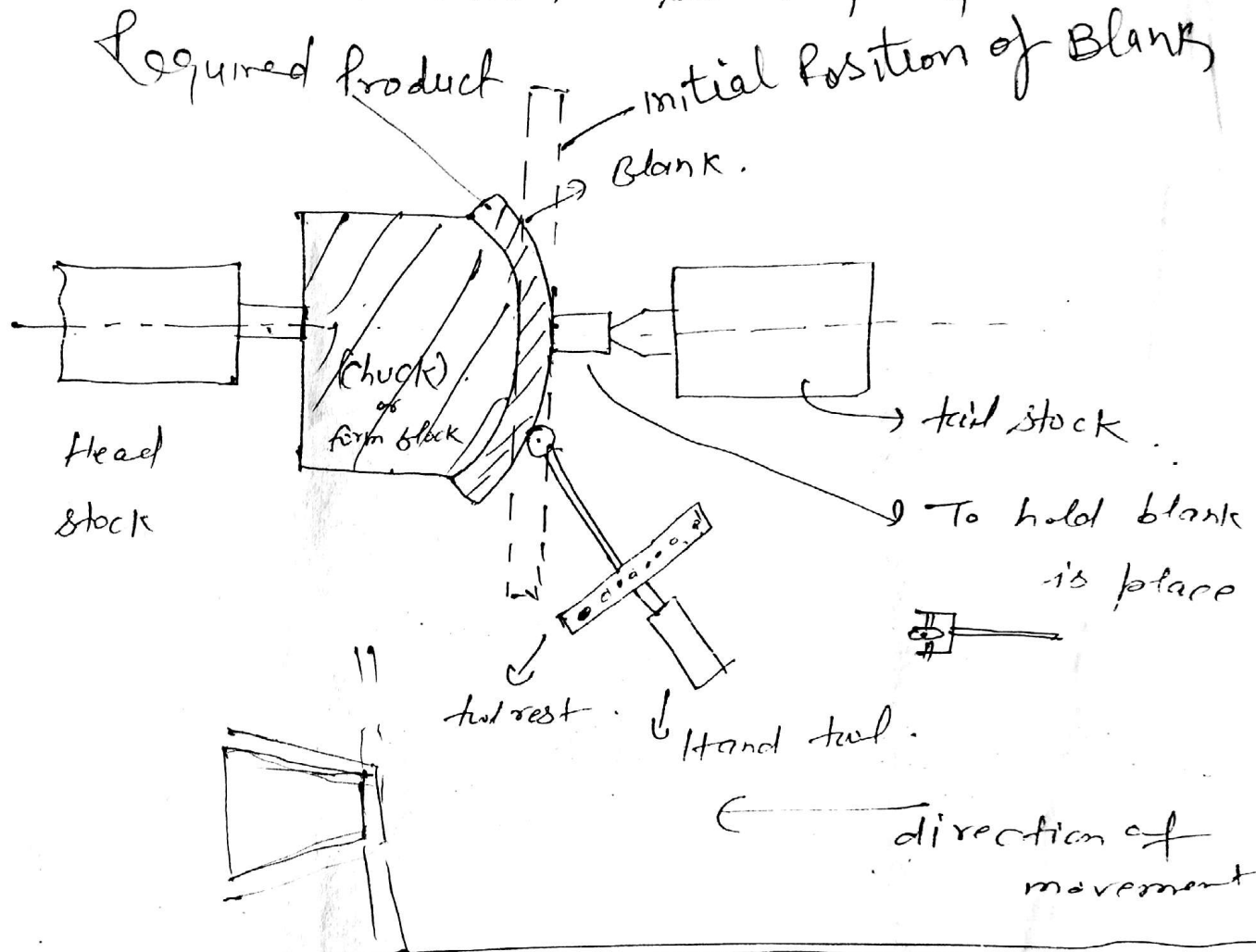
(a) Edge bending



(V-bending)

- Spinning → It is the process used for making cup & shaped parts having symmetrical shapes.
- In this, a blank is rotated, fixed against the form block and then gradually force is applied on the blank so that blank take the shape of form block.
- The process may be carried out on machine similar to a lathe machine.
- A circular blank properly centered with lathe axis is held against the chuck by the pressure of follower attached to tail stock.
- The tool used in spinning is simple round ended, wooden.

As the blank and form block are made to rotate, the tool is pressed and moved gradually on the blank so that blank takes the same shape of form block.



Stretch forming → In this, sheet metal is stretched

by two or more set of jaws that and simultaneously bent over a forming block or die.

- ↳ The material is stressed beyond the elastic limit and into plastic range.
- ↳ All the fibre of sheet material are stretched. Those on the inside of the radius of bend are.

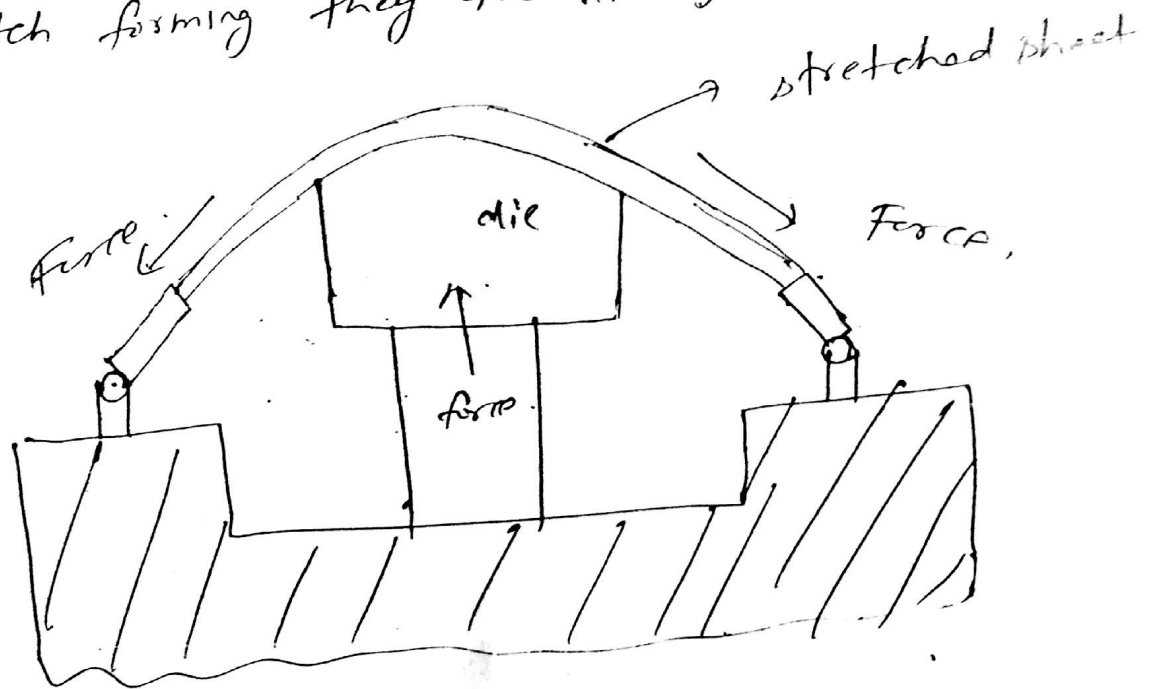
stretched less than those on the outside.

(13)

most of deformation is induced by the tensile stretching, therefore, force on the die are far less than those normally present in bending. Force are so low that die can often be made of wood or plastic.

The sheet should have uniform thickness otherwise the thinner portion are to be over stretched.

Also if sheet is having any hole before the stretch forming they are likely to be enlarged.

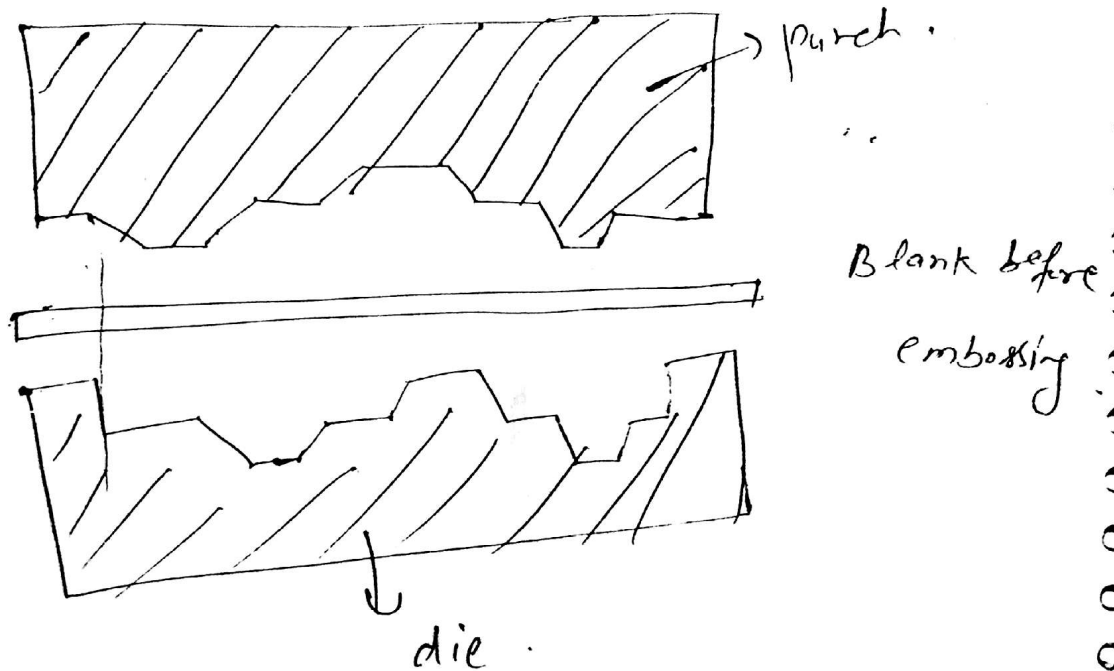


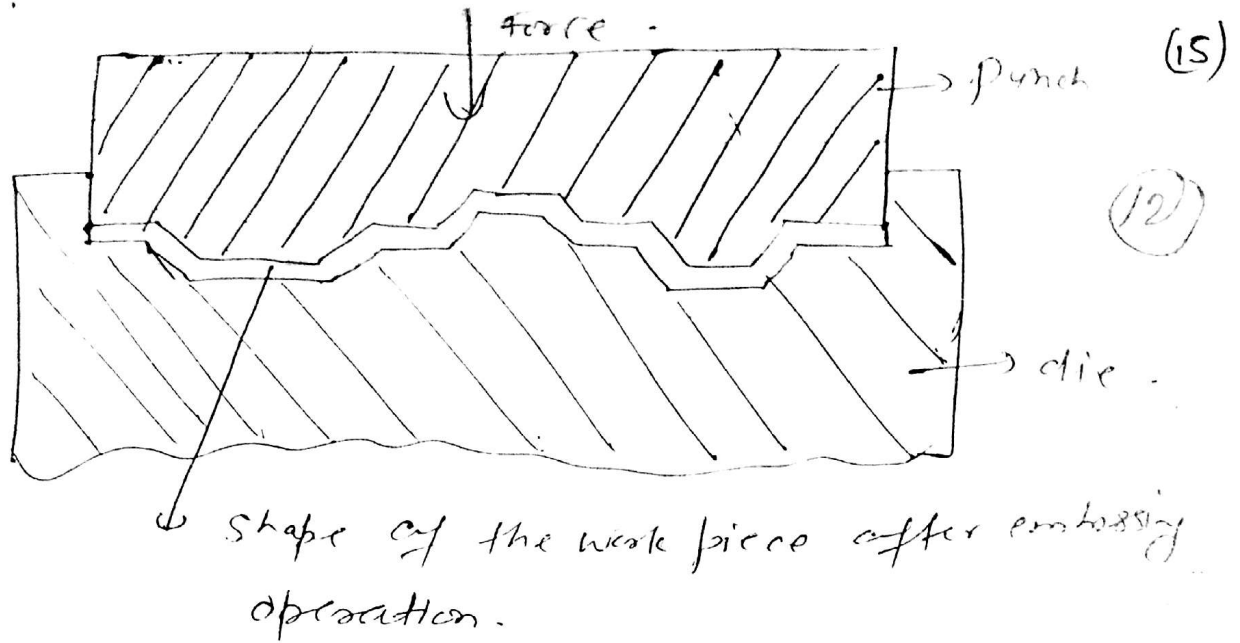
applications used in air-craft Industry for wing tip, scoop etc.

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Embossing → is the operation in which projected & recessed figure are made on sheet metal with corresponding relief on the other side.

- ↳ It involves drawing & bending operation. There may be negligible change in the thickness of the metal.
 - ↳ The die set consist of punch & die with required contours which are desired on the final product.
- The clearance b/w punch & die at meeting is same as the thickness of required product.
- ↳ It is used for providing rigidity to sheet metal and for decorative sheet work used in houses.

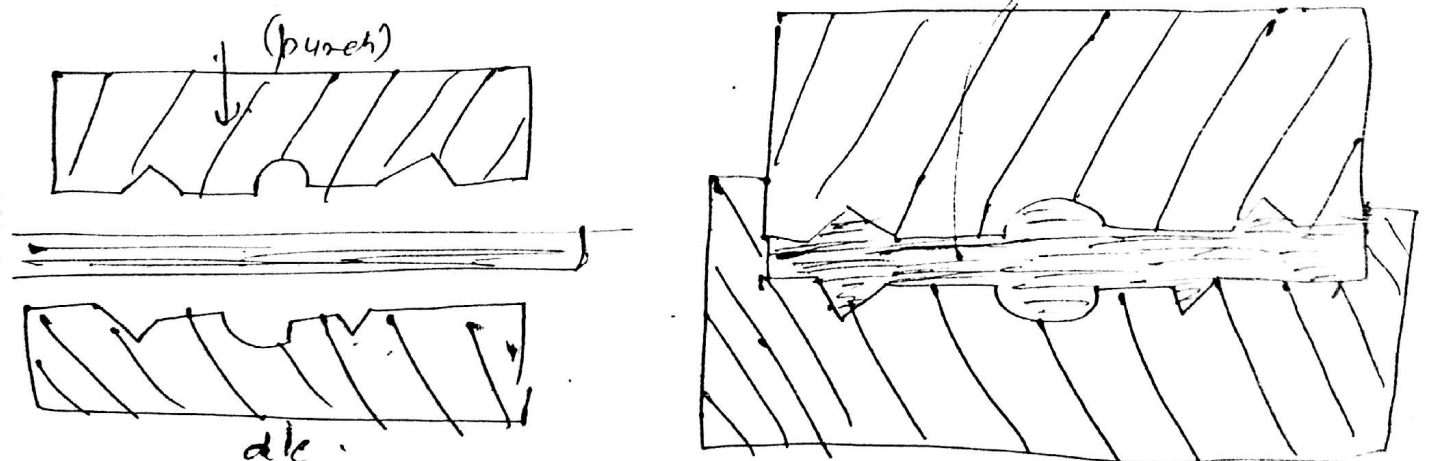




Coining → is a cold forging operation. Flow of metal occurs at top & bottom layer & not at the entire volume.

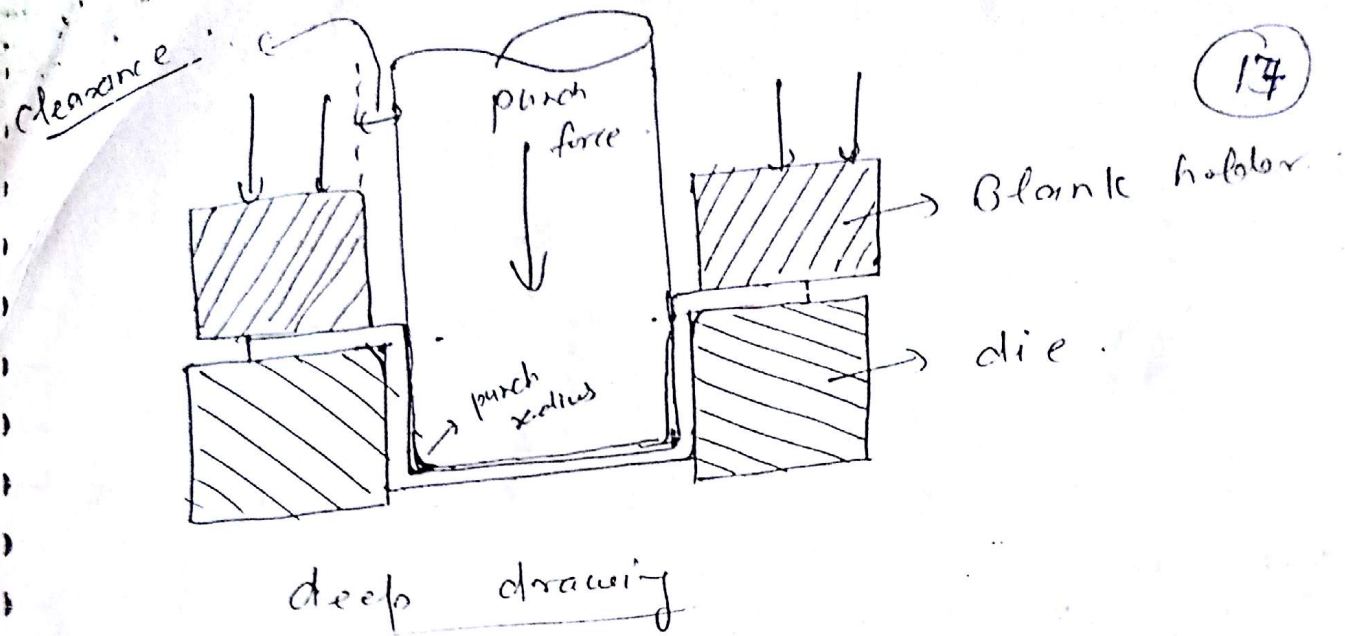
→ The process is used to produce coins, medals & similar product where exact size & fine details are required.

→ The coining die consist of punch & die which are engraved with necessary details required on the both side of final product. A blank is kept on the die & squeezed by it.



Sheet metal drawing $\rightarrow$ is a process for making cup shaped articles from flat sheet metal blank.

- ↳ Common examples dishes, Trays, brake drums, container etc
- ↳ punch & die are used with necessary rounding at the corner. This is provided to allow the smooth flow of metal during drawing.
- ↳ The blank is first placed & located above the die. The blank holder comes down & properly hold the edge of the blank
- ↳ After this, punch moves downwards to force the blank to take the shape of cup formed by the end of the punch. The punch & blank holder then returns upward to complete the cycle.
- ↳ The punch & die radius should be equal & sufficient large (about 5 times the thickness of metal). Large punch radius prevent fracture of metal near the end of punch where tensile forces are maximum.



Sheet metal joining

- 1) Rivetting
- 2) Screwing
- 3) Soldering, welding & Brazing
- 4) Folded sheet metal joints

Folded sheet metal joints

- ↳ Seam → joint on the sheet metal pieces through folding of edge is called seam joint.

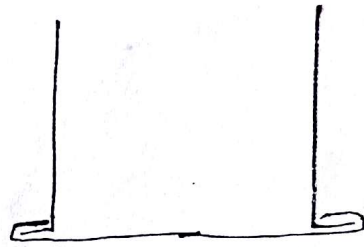
Type

- Grooved (Locked) Seam → A grooved seam is made by hooking two folded edge together & then off setting them. The joint is self locking.



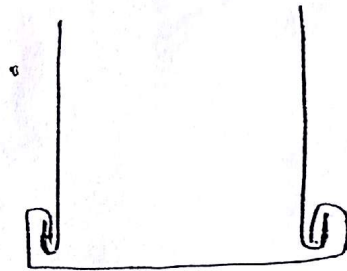
Grooved (locked) seam.

- 2) Single Seam → The seam is used to join a bottom portion to a vertical body. The bottom edge is hooked over the bent edge of the vertical body.



(Single seam).

- (3) Double Seam → The formed edge is bent upwards against the body.



double seam.

Edge firming or hemming →

- ↳ For sheet metal object strength is given to the edge and sharpness is eliminated by folding the edge, it is known as hemming.

types ① Single hem.

② Double hem.

③ Wired edge.


(single hem)

(a)


(double hem)

(a)

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(wire edge hem)

(c)

Wire edge hem → consist of an edge wrapped
around a steel wire for better strength.

* Sheet & metal work tool → done in workshop